Supervyse Storage System RAID Service D-bus API

Revision 1.7

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2017-12-15	Conrad Yao		Initial version
1.3	2020-02-15	Sam Wu	Edwin Liu	Add more APIs
1.4	2020-06-22	Sam Wu	Edwin Liu	Add SMART info
1.5	2023-06-20	Derick Lai	Edwin Liu	Add D-bus command example, such as Add JBOD, Add Copyback.
1.6	2024-10-21	Derick Lai	Nick Lin	Add D-bus command example
1.7	2024-12-10	Derick Lai	Nick Lin	Add RAID functions for Microchip mctp over PCIe

1 Terminology

Table 1-1

Term	Description
RAID	Redundant Array of Independent Disks.
HBA	Host Bus Adapters.
OOB	Out-of-Band.
B	This is only available for Broadcom RAID cards.
M	This is only available for Microchip RAID cards.
B/M	This is available for both Broadcom and Microchip RAID cards.
PMC	PMC-Sierra was a global fabless semiconductor company.
PBSI	PMC BMC Service Interface.
URI	Universal Resource Identifier
BBU	Battery Backup Module.
PCI	Peripheral Component Interconnect.
PCIe	PCI Express.
AHCI	Advanced Host Controller Interface.
UHCI	Universal Host Controller Interface.
SCSI	Small Computer System Interface.
SAS	Serial Attached SCSI.
SATA	Serial Advanced Technology Attachment.
NVM	Non-Volatile Memory.
NVMe	Non-Volatile Memory Express.
iSCSI	Internet SCSI.
RPM	Revolution(s) Per Minute.
JBOD	Just a Bunch of Drives.
HDD	Hard Disk Drive.
SSD	Solid State Disk.
SMART	Self-Monitoring Analysis and Reporting Technology.
BGI	Background Initialization.
aka	Also known as.
ZMR	From Microchip's Answer: ZMR mode is only available in very old products, you don't need to mention it.

2 Introduction

This document introduces Supervyse OPF Storage System RAID Service D-bus API and lists related features of RAID Service.

There are Broadcom and Microchip RAID/HBA cards supported. In the "Vendors Only" column of subsequent chapters, Broadcom RAID card and Microchip RAID card are abbreviated as "B" and "M" respectively. There are two OOB modes: I2C and MCTP over PCIe. The PBSI of Microchip is I2C base.

One of the most important functions is to create a virtual drive on a RAID card. A virtual drive is a partition in a drive group that is made up of contiguous data segments on the drives. Also called logical drive or volume. In the documentation, if you see virtual drive, logical drive or volume, they mean the same thing.

D-bus API is divided into seven categories: StorageCollection, Storage, StorageController, StorageControllerEvent, Drive, VolumeCollection and Volume. Their detailed descriptions are introduced in the subsequent chapters. In the object path of D-bus API, you will see the keywords <SYSTEMS_SN>, <STORAGE_ID> and <STORAGECONTROLLER_ID>. <SYSTEMS_SN> is "System Serial Number" used on Redfish URI, please refer to section "Get Redfish System URI SN" in document "Supervyse OPF IPMI OEM Command Specification". For <STORAGE_ID> and <STORAGECONTROLLER_ID>, please refer to Storage property "Id" and Controller property "Index" respectively in section "StorageSystem.xml" in document "Supervyse OPF Platform Porting Guide".

The following table lists the features of RAID Service. If the feature is supported, we marked as "V" or "A". Otherwise, we marked as "X". If marked as "V", you can find the corresponding d-bus command in the following chapters. If marked as "A", there is no such d-bus command because the feature is done by the RAID controller automatically.

Table 2-1

Function Name	Description	Broadcom(I2C/MCTP over PCIe)		Microchip (MCTP over PCIe)		Microchip (PBSI - I2C based)	
		RAID	HBA	RAID	HBA	RAID	HBA
Get Controller Information	Retrieve related information of RAID/HBA controller. Please refer to chapter "5.1" for more detail	V	V	V	V	V	V
Get Physical Drive Information	Retrieve related information of specific physical drive. Please refer to chapter "7.1", "7.2" and "7.3" for more detail.	V	V	V	V	V	V
Get Event Sequence Information	Retrieve total counts of events of RAID/HBA controller. Please refer to chapter "6.1" for more detail.	V	V	V	X	X	X
Get Event Details	Retrieve detail of each event of RAID/HBA controller. Please refer to chapter "6.2" for more detail.	V	V	V	X	X	X
Get Logical Drive Information	Retrieve related information of specific logical drive. Please refer to chapter "8.1" and "9.1" for more detail.	V	X	V	X	V	X
Create Logical Drive	Create a logical drive with specific physical drive. Two stages to create logical drive: "CreateVolumeBasicData" and "CreateVolumeParameters". Please refer to chapter "5.2" for more detail.	V	X	V	X	X	X
Delete Logical Drive	Remove specific existing logical drive. Please refer to chapter "5.2" for more detail.	V	X	V	X	X	X

Function Name	Description	Broadcom(I2C/MCTP over PCIe)		Microchip (MCTP over PCIe)		Microchip (PBSI - I2C based)	
		RAID	HBA	RAID	HBA	RAID	HBA
Clear Configuration	Remove all existing logical drives. Please refer to chapter "5.2" for more detail.	V	X	V	X	X	X
Clear Events	Clear all events. Please refer to chapter "5.2" for more detail.	V	X	X	X	X	X
Set Controller Properties	Set some properties of the controller. For Broadcom RAID, supports "Copyback". For Microchip RAID, supports "No Battery Write Cache", "Write Cache Policy" and "Background Consistency Check". Please refer to chapter "5.4" for more detail.	V	X	V	X	X	X
Set Logical Drive Properties	Set some properties of the logical drive. For Broadcom RAID, supports "Rename", "Read Policy", "Write Policy", "IO Policy", "Disk Cache Policy" and "BGI". For Microchip RAID, supports "Accelerator". Please refer to chapter "9.4" for more detail.	V	X	V	X	X	X
Get TTY Log	Get a copy of the firmware's terminal log entries for the controller. Please refer to chapter "5.2" for more detail.	V	X	V	X	X	X
Scan Foreign Configuration	Check whether the Controller has a foreign configuration. A **foreign configuration** is a RAID configuration that already exists on a replacement set of physical disks that you install in a computer system. Please refer to chapter "5.2" for more detail.	V	X	A	X	X	X
Import Foreign Configuration	Import foreign configuration to create a Logical Volume. Please refer to chapter "5.2" for more detail.	V	X	A	X	X	X
Clear Foreign Configuration	Clear existed foreign configuration. Please refer to chapter "5.2" for more detail.	V	X	A	X	X	X
Flash Controller Firmware	Firmware update for RAID Controller. Please refer to chapter "5.2" for more detail.	V	X	V	X	X	X
Start Patrol Read Operation	Trigger patrol read for current RAID Controller. **Patrol read** reviews your system for possible drive errors that could lead to drive failure and then performs action to correct errors. The goal is to protect data integrity by detecting drive failure before the failure can damage data. The corrective actions depend upon the drive group configuration and the type of errors. Please refer to chapter "5.2" for more detail.	V	X	X	X	X	X
Stop Patrol Read Operation	Cancel patrol read for current RAID Controller. Please refer to chapter "5.2" for more detail.	V	X	X	X	X	X
Make Dedicated Hot spare	Assign a Physical Drive to be a hot spare for a dedicated virtual drive. **Hotspare**: A drive that is powered up and ready for use as a spare in case an online drive fails. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X
Make Global Hot spare	Assign a Physical Drive to be a global hot spare drive for all virtual drives. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X
Remove Hot spare	Delete an existed hot spare drive. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X

Function Name	Description	Broadcom(I2C/MCTP over PCIe)		Microchip (MCTP over PCIe)		Microchip (PBSI - I2C based)	
		RAID	HBA	RAID	HBA	RAID	HBA
Locate Physical Drive	Locate a drive and activate the physical disk activity LED. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X
Stop Locate Physical Drive	Stops a locate operation and deactivates the drive's LED. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X
Start Replace physical drive	Replace a physical drive with another one. **Replace**: The Replace operation lets you copy data from a source drive into a destination drive that is not a part of the virtual drive. Please refer to chapter "7.4" for more detail.	V	X	V	X	X	X
Stop Replace physical drive	Stop to replace a physical drive with another one. Please refer to chapter "7.4" for more detail.	V	X	X	X	X	X
Check Consistency	Check the consistency of a specific logical drive. The **consistency check** operation verifies the data is correct in virtual drives that use RAID levels 1, 5, 6, 10, 50, and 60. RAID 0 does not provide data redundancy. For example, in a system with parity, checking consistency means calculating the data on one drive and comparing the results to the contents of the parity drive. For Microchip RAID, please refer to "Background Consistency Check". Please refer to chapter "9.2" for more detail.	V	X	V	X	X	X
Cancel Check Consistency	Stop checking the consistency of a specific logical drive. For Microchip RAID, please refer to "Background Consistency Check". Please refer to chapter "9.2" for more detail.	V	X	V	X	X	X
Locate Logical Drive	locate the physical drives in a virtual drive by flashing their LEDs. Please refer to chapter "9.2" for more detail.	V	X	X	X	X	X
Stop Locate Logical Drive	Stop locating the physical drives in a virtual drive and the LEDs on the physical drives stop flashing. Please refer to chapter "9.2" for more detail.	V	X	X	X	X	X
Perform Reconstruction	Change the RAID level of the specific logical drive. Please refer to chapter "9.2" for more detail.	V	X	V	X	X	X
Change Personality Mode	Switch between different personality modes (RAID or JBOD Mode) of the controller. Please refer to chapter "5.2" for more detail.	V	X	X	X	X	X

3 StorageCollection

StorageCollection is a group of storages in the RAID service. It contains storage count and IDs.

- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage
- Interface: com.insyde.storage.StorageCollection

3.1 Properties

Table 3.1-1

Name	Vendor Only	D-bus Type	Description
Description	B/M	STRING	"Storage System".
Name	B/M	STRING	"Insyde storage System".
Count	B/M	INT32	The storage count.
Members	B/M	ARRAY<STRING>	The storage IDs.

3.2 Method

There is no such implementation.

3.3 Signal

There is no such implementation.

3.4 Example

- Get StorageCollection Information

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageCollection"

Return result:
   array [
      dict entry(
         string "Description"
         variant          string "Storage System"
      )
      dict entry(
         string "Name"
         variant          string "Insyde Storage System"
      )
      dict entry(
         string "Count"
         variant          int32 1
      )
      dict entry(
         string "Members"
         variant          array [
               string "1"
```

```
        ]
    )
    dict entry(
      string "Result"
      variant          string "Success"
    )
  ]
```

4 Storage

Storage is a group that contains a set of storage controllers and drives.
- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>
- Interface: com.insyde.storage.Storage

4.1 Properties

Table 4.1-1

Name	Vendor Only	D-bus Type	Description
Description	B/M	STRING	Description of Storage.
Name	B/M	STRING	Name of Storage.
StorageControllersCount	B/M	INT32	The StorageController count.
StorageControllers	B/M	ARRAY<STRING>	StorageController IDs.
StorageDrivesCount	B/M	INT32	The Drive count.
StorageDrives	B/M	ARRAY<STRING>	The Drive IDs.
Status.State	B/M	STRING	The state of storage device. • **Enabled**, Normal. • **Disabled**, Initialization failure.
Status.Health	B/M	STRING	The health state of this resource is in the absence of its dependent resources. • **OK**, Normal.
Status.HealthRollup	B/M	STRING	This represents the overall health state from the view of this resource. • **OK**, Normal.

4.2 Method

There is no such implementation.

4.3 Signal

There is no such implementation.

4.4 Example

- Get Storage Information

```
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789abcdef/Storage/1
org.freedesktop.DBus.Properties.GetAll string:"com.insyde.storage.Storage"

Return result:
   array [
      dict entry(
         string "Description"
         variant          string "Storage System"
      )
      dict entry(
```

```
      string "Name"
      variant         string "Storage System"
    )
    dict entry(
      string "StorageDrivesCount"
      variant         int32 8
    )
    dict entry(
      string "StorageControllersCount"
      variant         int32 1
    )
    dict entry(
      string "StorageControllers"
      variant         array [
          string "1"
        ]
    )
    dict entry(
      string "StorageDrives"
      variant         array [
          string "0"
          string "1"
          string "2"
          string "3"
          string "4"
          string "5"
          string "6"
          string "7"
        ]
    )
    dict entry(
      string "Status.State"
      variant         string "Enabled"
    )
    dict entry(
      string "Status.Health"
      variant         string "OK"
    )
    dict entry(
      string "Status.HealthRollup"
      variant         string "OK"
    )
    dict entry(
      string "Result"
      variant         string "Success"
    )
  ]
```

5 StorageController

StorageController is a RAID/HBA card, which contains the firmware version, Manufacturer, Model and so on.

- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/ StorageControllers/<STORAGECONTROLLER_ID>
- Interface: com.insyde.storage.StorageController

5.1 Properties

Table 5.1-1

Name	Vendor Only	D-bus Type	Description
Name	B/M	STRING	The name of the StorageController.
Description	B/M	STRING	Description about the StorageController.
SpeedGbps	B/M	DOUBLE	The speed of the storage controller interface.
FirmwareVersion	B/M	STRING	The firmware version of this storage Controller.
Manufacturer	B/M	STRING	This is the manufacturer of this storage controller.
Model	B/M	STRING	This is the model number for the storage controller.
SerialNumber	B/M	STRING	The serial number for this storage controller.
SupportedControllerProtocols	B/M	ARRAY<STRING>	The protocol by which this storage controller can be communicated to. • PCIe (Vendor Proprietary).
SupportedDeviceProtocols	B/M	ARRAY<STRING>	This represents the protocols by which this storage controller can be communicated to. Valid Values: • SAS • SATA
Status.State	B/M	STRING	This indicates the known state of the resource, such as if it is enabled. Valid values for Broadcom: • **Enabled**, this function or resource has been enabled. • **Disabled**, this function or resource has been disabled. Valid values for Microchip: • **Ok** • **Failed** • **Bad ZMR**, the controller is disabled because the cache module is not attached. • **Cache Problem** • **Incompatible Cache**, incompatible cache module attached. • **Dirty Cache Requires Downgrade**, dirty cache exists and requires a firmware downgrade to flush.

Name	Vendor Only	D-bus Type	Description
Status.Health	B/M	STRING	This represents the health state of this resource in the absence of its dependent resources. Valid Values: • **OK**, Normal. • **Warning**, A condition exists that requires attention. • **Critical**, A critical condition exists that requires immediate attention.
Status.HealthRollup	B/M	STRING	This represents the overall health state from the view of this resource. Valid Values: • **OK**, Normal. • **Warning**, A condition exists that requires attention. • **Critical**, A critical condition exists that requires immediate attention.
Oem.BBUPresent	B/M	BOOL	The present of BBU module. If there is no BBU module, the property will not be chdisplayed.
Oem.BBUType	B/M	STRING	The type of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUName	B	STRING	The model of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUStatus	B/M	STRING	The status of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUTemperatureCelsius	B/M	INT32	The temperature of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUVoltageMilliVolts	B/M	UINT32	The Voltage(mV) of BBU module. The temperature of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUCurrentMilliAmps	B/M	INT32	The Current(mA) of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUSerialNumber	B	STRING	The serial number of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBUHealthPercent	B/M	INT32	The health percentage of BBU module. If there is no BBU module, the property will not be displayed.
Oem.BBURemainingBatteryPercent	B	INT32	The Remaining Battery percentage of BBU module. If there is no BBU module, the property will not be displayed.
Oem.JunctionTemperature	B/M	INT32	The junction temperature of controller.
Oem.Chip	B/M	STRING	The chip name of controller.
Oem.CacheIsPinned	B	STRING	If the controller loses access to one or more virtual drives, the controller preserves the data from the virtual drive. This preserved cache is called as pinned cache. Valid values: • **Yes** • **No**
Oem.BIOSVersion	B	STRING	The BIOS version of controller.
Oem.UEFIVersion	B	STRING	The UEFI version of controller.
Oem.PCIVendorID	B/M	STRING	PCI vendor ID of controller.

Name	Vendor Only	D-bus Type	Description
Oem.PCIDeviceID	B/M	STRING	PCI Device ID of controller.
Oem.PCISubVersionID	B/M	STRING	PCI Subversion ID of controller.
Oem.PCISubSystemID	B/M	STRING	PCI Subsystem ID of controller.
Oem.MemoryCorrectErrCount	B	INT32	The correctable error count of memory.
Oem.MemoryUnCorrectErrCount	B	INT32	The uncorrectable error count of memory.
Oem.HostInterface	B/M	STRING	Host interface information of controller.
Oem.PortCount	B	INT32	The count of ports for SAS/iSCSI.
Link.Oem.StorageDrives	B/M	ARRAY<STRING>	Contains references to Drives that are related to this resource.
Link.Oem.StorageVolumes	B/M	ARRAY<STRING>	Contains references to Volumes that are related to this resource.
Oem.CopyBackDisabled	B	BOOLEAN	Copyback operation. ● true: Disable CopyBack ● flase: Enable CopyBack Description: When a drive fails or is expected to fail, the data is rebuilt on a hot spare. The failed drive is replaced with a new disk. Then the data is copied from the online drive (which was previously an hot spare) to the new drive, and the hot spare reverts from a rebuilt drive to its original hot spare status. Replace runs as a background activity, and the virtual drive is still available online to the host.
Oem.CopybackOnSmartErrorEnabled	B	BOOLEAN	Copyback on SMART error. ● true: enable ● false: disable Description: Replace also is initiated when the first SMART error occurs on a drive that is part of a virtual drive. The destination drive is a hot spare that qualifies as a rebuild drive. The drive that has the SMART error is marked as failed only after the successful completion of Replace. This situation avoids putting the drive group in Degraded status.
Oem.CopybackToSSDOnSmartErrorEnabled	B	BOOLEAN	Copyback to SSD on SMART error. ● true: enable ● false: disable
Oem.NoBatteryWriteCache	M	STRING	No Battery Write Cache. ● Enabled ● Disabled ● Invalid (Ex. No logical drive) Description: Allows the controller to enable write cache when no battery is present or when the battery fails. Values are Enable or Disable. Enabling No-Battery Write Cache can result in data loss if the server loses power and there is no battery present or the battery has failed.

Name	Vendor Only	D-bus Type	Description
Oem.DriveWriteCachePolicy	M	STRING	Drive Write Cache Policy. ● Default ● Enabled ● Disabled ● Unchanged Descritption: ● Default: Parameter to have a default or no policy for physical drive write cache. ○ For drives with usage type configured, this policy means that the controller firmware will enable and disable the drive write cache on each physical drive to maximize efficiency and data protection. ○ For drives with usage type unconfigured and HBA, this policy means that the controller firmware will not modify the drive write cache of the physical drive configured via other tools or utilities. ● Enable: Parameter to enable the physical drive write cache. => This policy means that the controller will enable the drive write cache on the selected physical drives. ● Disable: Parameter to disable the physical drive write cache. => This policy means that the controller will disable the drive write cache on the selected physical drives. ● Unchanged: Parameter to leave the physical drive write cache unchanged. => This policy means that the controller will not modify the drive write cache on the selected physical drives.
Oem.SurfaceScanAnalysisPriority	M	STRING	The priority of surface scan analysis（background consistency check）. Valid Value: ● Disabled ● High ● Idle ● Invalid Description: Surface Scan Analysis is also called "Background Consistency Check", controller firmware continually and automatically checks your logical drives for bad or inconsistent data when they are in use. ● Disabled: The consistency check is disabled. ● High: The background consistency check runs in parallel to host I/O and might have an impact on performance.

Name	Vendor Only	D-bus Type	Description
			● Idle: The background consistency check only runs during periods of host inactivity and does not impact performance. ● Invalid: No logical drive.
Oem.SurfaceScanAnalysis Delay	M	UINT16	Total amount of controller delay time (in tenths-of-seconds) that must elapse before background surface scan analysis (background consistency check) activity begins on any volume. The property shows when the value of "Oem.SurfaceScanAnalysisPriority" is Idle. Valid value: 0 - 300. ● 0: The Oem.SurfaceScanAnalysisPriority is Disabled. ● 0: The Oem.SurfaceScanAnalysisPriority is High. ● 2-300: The Oem.SurfaceScanAnalysisPriority is Idle.

5.2 Method

Table 5.2-1

Name	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
GetFreeRaidResource	B/M	Get free raid resource includes free drives and possible RAID Level.	N/A	• ARRAY ○ STRING <FreeDriveIds> • ARRAY ○ STRING <PossibleRaidLevel> Ex. □ 0 □ 1 □ 5 □ 6 □ 10 □ 50 □ 60 • STRING ○ Success ○ Failure
CreateVolumeBasicData	B	First stage to create logical drive. Use RAID level and the IDs of drives to get maximum logical drive size, stripe size, access policy, disk cache policy and init state. For AccessPolicy, please read the description of Oem.AccessPolicy. For DiskCachePolicy, please read the description of Oem.DiskCachePolicy.	• STRING RaidLevel • ARRAY ○ STRING SelectedDriveId	• UINT32 <MaxSize (Byte)> • UINT32 <StripeSize (KB)> • ARRAY <STRING <AccessPolicy>>> ○ **Read Write** ○ **Read Only** ○ **Blocked** • ARRAY ○ STRING <DiskCachePolicy> □ **Unchanged** □ **Enable** □ **Disable** • ARRAY ○ STRING <InitState> □ **No Init**, Do not initialize the virtual drive. □ **Quick Init**, Initializes the first 8 MB and the last 8 MB on the virtual drive. □ **Full Init**, Initializes the entire virtual drive. • STRING ○ Success ○ Failure
CreateVolumeBasicData	M	First stage to create logical drive. Use RAID level and the IDs of drives to get	• STRING RaidLevel • ARRAY ○ STRING SelectedDriveId	• UINT32 <MaxSize (Byte)> • UINT32 <StripeSize (KB)> • ARRAY ○ STRING <Accelerator> □ **Controller Cache** □ **None**

Name	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
		maximum logical drive size, stripe size, accelerator and init method. For Accelerator, please read the description of Oem.Accelerator. For InitMethod, please read the desctription of Oem.InitMethod.		● ARRAY ○ STRING <InitMethod> □ **Default** □ **Rapid Parity Init** (For RAID5,6,50 and 60) ● STRING ○ Success ○ Failure
CreateVolume Parametere	B	Second stage to create logical drive.	● STRING Logical Drive Name ● UINT32 MaxSize ● UINT32 StripeSize (KB) ● STRING AccessPolicy Ex. Read Write Read Only Blocked ● STRING DiskCachePolicy Ex. Unchanged Enable Disable ● STRING InitState Ex. No Init Quick Init Full Init	● STRING ○ Success ○ Failure
CreateVolume Parameters	M	Second stage to create logical drive.	● STRING LogicalDriveName ● UINT32 MaxSize ● UINT32 StripeSize (KB) ● STRING Accelerator Ex. Controller Cache None ● STRING InitMethod Ex. Default Rapid Parity Init	● STRING ○ Success ○ Failure

Name	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
			• STRING Reserve Ex. Reserve	
DeleteVolume	B/M	Delete a volume with specific ID.	• STRING VolumeID	• STRING ○ Success ○ Failure
SaveTTYLog	B/M	Save TTY Log.	N/A	• ARRAY ○ STRING Logs • STRING ○ Success ○ Failure
ScanForeignConfiguration	B	Use Scan Foreign Configuration to find drives with foreign configurations.	N/A	• STRING TargetId • STRING RaidLevel • UINT32 Size(MB) • STRING State ○ **OPTIMAL** - A virtual drive whose members are all online. ○ **PARTIALLY DEGRADED** - A virtual drive with a redundant RAID level that is capable of sustaining more than one member drive failure. This state also applies to the virtual drive's member drives. Currently, a RAID 6 or RAID 60 virtual drive is the only virtual drive that can be partially degraded. ○ **DEGRADED** - A virtual drive with a redundant RAID level with one or more member failures and can no longer sustain a subsequent drive failure. ○ **OFFLINE** - A virtual drive with one or more member failures that make the data inaccessible. • STRING ○ Success ○ Failure
ImportForeignConfiguration	B	Import the existing configuration to the RAID controller.	N/A	• STRING ○ Success ○ Failure
ClearForeignConfiguration	B	Clear Foreign Configuration.	N/A	• STRING ○ Success ○ Failure
FirmwareUpdateStart	B	Start Firmware Upddate.	• STRING <location of firmware image>	• STRING ○ Success ○ Failure

Name	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
FirmwareUpdateGetState	B	Get current state of Firmware Upddate.	N/A	• STRING ○ Idle ○ Running ○ Success ○ Failed
StartPatrolRead	B	Start Patrol Read for Physical Drives.	N/A	• STRING ○ Success ○ Failure
StopPatrolRead	B	Stop Patrol Read for Physical Drives.	N/A	• STRING ○ Success ○ Failure
ClearConfiguration	B/M	Delete all volumes.	N/A	• STRING ○ Success ○ Failure
ClearEvents	B	Clear all events.	N/A	• STRING ○ Success ○ Failure
SetFactoryDefaults	B	Reset all controller properties and the volumes will be preserved. There will be no change to the config.	N/A	• INT32 return_value ○ 116: Success and reboot required ○ Other value: Failure • STRING ○ Success ○ Failure
GetPersonality	B	Get personality mode. The MegaRAID firmware supports two personality modes: RAID and JBOD.	N/A	• STRING current mode ○ RAID ○ JBOD • STRING requested mode ○ RAID ○ JBOD ○ NULL (no requested mode) • STRING ○ Success ○ Failure
SetPersonality	B	Set personality mode: RAID or JBOD. Note_1: Should not have any volume before setting personality mode. Note_2: The possible RAID levels in JBOD mode are RAID0, 1, 10.	• UINT32 <mode> Ex. 1: RAID mode 4: JBOD mode	• INT32 return_value ○ 116 : Success and reboot required ○ Other value : Failure • STRING ○ Success ○ Failure

5.3 Signal

There is no such implementation.

5.4 Property Set

- Object Path: /com/insyde/storage/Systems/<SYSETMS_ID>/Storage/<STORAGE_ID>/StorageControllers/<STORAGECONTROLLER_ID>
- Interface: org.freedesktop.DBus.Properties
- Method: Set

Table 5.4-1

Feature	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
Copyback	B	Broadcom has changed Copyback operation name to Replace on June 17, 2021. Replace lets you copy data from a source drive to a destination drive that is not a part of the virtual drive. Typically, when a drive fails or is expected to fail, the data is rebuilt on a hot spare. The failed drive is replaced with a new disk. Then the data is copied from the online drive (which was previously an hot spare) to the new drive, and the hot spare reverts from a rebuilt drive to its original hot spare status. Replace runs as a background activity, and the virtual drive is still available online to the host. Replace also is initiated when the first SMART error occurs on a drive that is part of a virtual drive. The destination drive is a hot spare that qualifies as a rebuild drive. The drive that has the SMART error is marked as failed only after the successful completion of Replace. This situation avoids putting the drive group in Degraded status.	• STRING "com.insyde.storage.StorageController" • STRING "CopyBackDisabled" • BOOLEAN <true/false> Ex. **true**: Disable Copyback. **false**: Enable Copyback. • STRING "Copyback on SMART error Enabled" • BOOLEAN <true/false> Ex. **true**: Enable Copyback on SMART error. **false**: Disable Copyback on SMART error. • STRING "Copyback to SSD on SMART error Enabled" • BOOLEAN <true/false> Ex. **true**: Enable Copyback to SSD on SMART error. **false**: Disable Copyback to SSD on SMART error.	N/A
No Battery Write Cache	M	No-Battery Write Cache setting decides the controller to enable write cache when no battery is present or when	• STRING "com.insyde.storage.StorageController" • STRING "MC No Battery Write Cache" • STRING <Value> Ex.	N/A

Feature	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
		the battery fails. Values are Enable or Disable.	Enabled Disabled	
Write Cache Policy	M	Write Cache Policy Default: For drives with usage type configured, this policy means that the controller firmware will enable and disable the drive write cache on each physical drive to maximize efficiency and data protection. For drives with usage type unconfigured or HBA, this policy means that the controller firmware will not modify the drive write cache of the physical drive configured via other tools or utilities. Enabled: This policy means that the controller will enable the drive write cache on the selected physical drives. Disabled: This policy means that the controller will disable the drive write cache on the selected physical drives. Unchanged: This policy means that the controller will not modify the drive write cache on the selected physical drives.	• STRING "com.insyde.storage.StorageController" • STRING "MC Write Cache Policy" • STRING <Value> Ex. Default Enabled Disabled Unchanged	N/A
Surface Scan Analysis (Background Consistency Check)	M	If your controller supports background consistency check, controller firmware continually and automatically checks your logical drives for bad or inconsistent data when they are in use. When enabled, Consistency Check performs a background check on logical drives every 14 days from the time the last check was completed. Turns background consistency check on, with optional 0 - 300 tenths of second delay period. The delay period sets the controller idle time, after which the consistency check will start. A value of 0 disables the consistency check. If the delay is 1, the consistency check mode is set to HIGH. If the delay is 2 – 300, the consistency check mode is set to Idle. When set to High, the check runs in parallel to host I/O and might have an impact on performance. When set to idle, the check only runs during periods of host inactivity and does not impact performance.	• STRING "com.insyde.storage.StorageController" • STRING "MC SSA Delay" • UINT16 <Time in tenths of second> Ex. 0: Disabled 1: High 2 – 300: Idle > 300: Invalid	N/A

5.5 Example

- Get Controller Information

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1  org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageController"

Return result:
   array [
      dict entry(
         string "Name"
         variant          string "Broadcom"
      )
      dict entry(
         string "Description"
         variant          string "Broadcom RAID/HBA Controller"
      )
      dict entry(
         string "SpeedGbps"
         variant          double 12
      )
      dict entry(
         string "FirmwareVersion"
         variant          string "5.220.02-3691"
      )
      dict entry(
         string "PackageVersion"
         variant          string "52.22.0-4544"
      )
      dict entry(
         string "Manufacturer"
         variant          string "Broadcom"
      )
      dict entry(
         string "Model"
         variant          string "MegaRAID 9560-16i 4GB"
      )
...
```

- GetFreeRaidResource

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.GetFreeRaidResource

Return result:
   array [
      string "0"
      string "1"
      string "2"
      string "3"
      string "4"
      string "5"
      string "6"
      string "7"
```

```
  ]
  array [
    string "0"
    string "1"
    string "5"
    string "6"
    string "10"
    string "50"
    string "60"
  ]
  array [
    dict entry(
      string "Result"
      variant         string "Success"
    )
  ]
```

- Create Logical Drive - For Broadcom RAID
  - Stage1: CreateVolumeBasicData - For Broadcom RAID

```
# dbus-send --system --print-reply --reply-timeout=30000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.CreateVolumeBasicData string:"1" array:string:"0","1"

Return result:
   array [
    uint32 64
    uint32 128
    uint32 256
    uint32 512
    uint32 1024
  ]
  array [
    string "Read Write"
    string "Read Only"
    string "Blocked"
  ]
  array [
    string "Unchanged"
    string "Enable"
    string "Disable"
  ]
  array [
    string "No Init"
    string "Quick Init"
    string "Full Init"
  ]
  array [
    dict entry(
      string "Result"
      variant         string "Success"
    )
  ]
```

  - Stage2: CreateVolumeParameters - For Broadcom RAID

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.CreateVolumeParameters string:"Test" uint32:237952 uint32:128 string:"Read Write"
string:"Enable" string:"No Init"
```

- Create Logical Drive - For Microchip RAID (MCTP over PCIe)
  - Stage1: CreateVolumeBasicData - For Microchip RAID (MCTP over PCIe)

```
# dbus-send --system --print-reply --reply-timeout=30000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.CreateVolumeBasicData string:"1" array:string:"0","1"

Return result:
   array [
      uint32 16
      uint32 32
      uint32 64
      uint32 128
      uint32 256
      uint32 512
      uint32 1024
   ]
   array [
      string "Controller Cache"
      string "None"
   ]
   array [
      string "Default"
   ]
   array [
   ]
   array [
      dict entry(
         string "Result"
         variant          string "Success"
      )
   ]
```

  - Stage2: CreateVolumeParameters - For Microchip RAID (MCTP over PCIe)

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.CreateVolumeParameters string:"Test" uint32:476908 uint32:256 string:"None"
string:"Default" string:"Reserve"
```

- Delete Logical Drive

```
# dbus-send --system --print-reply --reply-timeout=30000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.DeleteVolume
string:"Broadcom0_1"
```

- Get StorageController Properties "Copyback" - For Broadcom RAID

```
ex.
dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1  org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageController"

Return result:
   array [
      ...
      dict entry(
         string "Oem.CopyBackDisabled"
         variant          boolean false
      )
      dict entry(
         string "Oem.CopybackOnSmartErrorEnabled"
```

```
      variant           boolean true
    )
    dict entry(
      string "Oem.CopybackToSSDOnSmartErrorEnabled"
      variant           boolean true
    )
    ...
  ]
```

- Set StorageController Properties "Copyback" - For Broadcom RAID
  - Oem.CopyBackDisabled
  - Oem.CopybackOnSmartErrorEnabled
  - Oem.CopybackToSSDOnSmartErrorEnabled

```
[Disable Copyback]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"CopyBackDisabled" variant:boolean:true string:"Copyback on SMART
error Enabled" variant:boolean:false string:"Copyback to SSD on SMART error Enabled" variant:boolean:false


[Enable Copyback]
[[CopyBackDisabled=false, CopybackOnSmartErrorEnabled=false, CopybackToSSDOnSmartErrorEnabled=false]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"CopyBackDisabled" variant:boolean:false string:"Copyback on SMART
error Enabled" variant:boolean:false string:"Copyback to SSD on SMART error Enabled" variant:boolean:false

[[CopyBackDisabled=false, CopybackOnSmartErrorEnabled=true, CopybackToSSDOnSmartErrorEnabled=false]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"CopyBackDisabled" variant:boolean:false string:"Copyback on SMART
error Enabled" variant:boolean:true string:"Copyback to SSD on SMART error Enabled" variant:boolean:false

[[CopyBackDisabled=false, CopybackOnSmartErrorEnabled=false, CopybackToSSDOnSmartErrorEnabled=true]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"CopyBackDisabled" variant:boolean:false string:"Copyback on SMART
error Enabled" variant:boolean:false string:"Copyback to SSD on SMART error Enabled" variant:boolean:true

[[CopyBackDisabled=false, CopybackOnSmartErrorEnabled=true, CopybackToSSDOnSmartErrorEnabled=true]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"CopyBackDisabled" variant:boolean:false string:"Copyback on SMART
error Enabled" variant:boolean:true string:"Copyback to SSD on SMART error Enabled" variant:boolean:true
```

- Get/Set StorageController Properties "No Battery Write Cache" - For Microchip RAID (MCTP over PCIe)
  - Oem.NoBatteryWriteCache

```
[Get]
dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1  org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageController"

Return result:
method return time=1666847758.349978 sender=:1.1153 -> destination=:1.1176 serial=49 reply_serial=2
   array [
      ...
      dict entry(
        string "Oem.NoBatteryWriteCache"
```

```
        variant          string "Enabled"
      )
      ...
   ]

[Set]
[[Enabled]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC No Battery Write Cache" variant:string:"Enabled"
[[Disabled]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC No Battery Write Cache" variant:string:"Disabled"
```

- Get/Set StorageController Properties "Write Cache Policy" - For Microchip RAID (MCTP over PCIe)
  - Oem.DriveWriteCachePolicy

```
[Get]
dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1  org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageController"

Return result:
   array [
      ...
      dict entry(
         string "Oem.DriveWriteCachePolicy"
         variant          string "Enabled"
      )
      ...
   ]

[Set]
[[Default]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC Write Cache Policy" variant:string:"Default"
[[Enabled]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC Write Cache Policy" variant:string:"Enabled"
[[Disabled]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC Write Cache Policy" variant:string:"Disabled"
[[Unchanged]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC Write Cache Policy" variant:string:"Unchanged"
```

- Get/Set StorageController Properties "Surface San Analysis" - For Microchip RAID (MCTP over PCIe)
  - Oem.SurfaceScanAnalysisPriority and Oem.SurfaceScanAnalysisDelay

```
[Get]
dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1  org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageController"
```

```
Return result:
[[Case1: No logical drives]]
  array [
    ...
    dict entry(
      string "Oem.SurfaceScanAnalysisPriority"
      variant          string "Invalid"
    )
    ...
  ]
[[Case2: Surface Scan Analysis Priority Disabled]]
  array [
    ...
    dict entry(
      string "Oem.SurfaceScanAnalysisPriority"
      variant          string "Disabled"
    )
    dict entry(
      string "Oem.SurfaceScanAnalysisDelay"
      variant          uint16 0
    )
    ...
  ]

[[Case3: Surface Scan Analysis Priority High]]
  array [
    ...
    dict entry(
      string "Oem.SurfaceScanAnalysisPriority"
      variant          string "High"
    )
    dict entry(
      string "Oem.SurfaceScanAnalysisDelay"
      variant          uint16 1
    )
    ...
  ]

[[Case4: Surface Scan Analysis Priority Idle]]
  array [
    ...
    dict entry(
      string "Oem.SurfaceScanAnalysisPriority"
      variant          string "Idle"
    )
    dict entry(
      string "Oem.SurfaceScanAnalysisDelay"
      variant          uint16 40
    )
    ...
  ]

[Set]
[[Set Surface Scan Analysis Priority Disabled]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC SSA Delay" variant:uint16:0
[[Set Surface Scan Analysis Priority High]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC SSA Delay" variant:uint16:0
```

```
[[Set Surface Scan Analysis Priority Idle]]
dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.StorageController" string:"MC SSA Delay" variant:uint16:30
```

● Patrol Read - For Broadcom RAID

```
[Start Patrol Read Operation]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.StartPatrolRead

[Stop Patrol Read Operation]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.StopPatrolRead
```

● Get TTY Log

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.SaveTTYLog

Return result:
   array [
      dict entry(
         string "Result"
         variant          string "Success"
      )
   ]
```

● Scan Foreign Configuration - For Broadcom RAID

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.ScanForeignConfiguration

Return result:
   string "0"
   string "1"
   uint32 476416
   string "OPTIMAL"
   array [
      dict entry(
         string "Result"
         variant          string "Success"
      )
   ]
```

● Import Foreign Configuration - For Broadcom RAID

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.ImportForeignConfiguration
```

● Clear Foreign Configuration - For Broadcom RAID

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.ClearForeignConfiguration
```

● Clear Configuration

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.ClearConfiguration
```

● Clear Events - For Broadcom RAID

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.ClearEvents
```

● Start Controller Firmware Update

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.FirmwareUpdateStart string:"/tmp/mr3108fw.rom"
```

● Get State of Controller Firmware Update

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.FirmwareUpdateGetState
Return result:
   array [
      dict entry(
         string "Result"
         variant          string "Running"
      )
   ]
```

● Factory Default - For Broadcom RAID

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.SetFactoryDefaults
```

● GetPersonality - For Broadcom RAID

```
# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.GetPersonality

Return Result:
Ex. Current mode JBOD; Requested mode RAID
   string "JBOD"
   string "RAID"
   array [
      dict entry(
         string "Result"
         variant          string "Success"
      )
   ]
   int32 0

Ex. Current mode RAID; No requested mode
   string "RAID"
   string ""
   array [
      dict entry(
         string "Result"
         variant          string "Success"
      )
   ]
   int32 0
```

● SetPersonality (Set RAID mode) - For Broadcom RAID

```
Environment: Should not have any volume before setting personality mode.

# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.SetPersonality
uint32:1

Return result:
   int32 116
```

```
   array [
      dict entry(
         string "Result"
         variant             string "Success"
      )
   ]
   int32 0
```

● SetPersonality (Set JBOD mode) - For Broadcom RAID

```
Environment: Should not have any volume before setting personality mode.

# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/StorageControllers/1 com.insyde.storage.StorageController.SetPersonality
uint32:4

Return result:
   int32 116
   array [
      dict entry(
         string "Result"
         variant             string "Success"
      )
   ]
   int32 0
```

6 StorageControllerEvent

StorageControllerEvent is the event messages received from the StorageController.

- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/ StorageControllers/<STORAGECONTROLLER_ID>/Events
- Interface: com.insyde.storage.StorageControllerEvent

6.1 Properties

Table 6.1-1

Name	Vendor only	D-bus Type	Description
SeqNumNewestEvent	B/M	UINT32	The newest sequential number of events. If there is no event function, the property will not be displayed.
SeqNumOldestEvent	B/M	UINT32	The oldest sequential number of events. If there is no event function, the property will not be displayed.
SeqNumLastClearEvent	B	UINT32	The last clear sequential number of events. If there is no event function, the property will not be displayed.
SeqNumLastCleanShutdownEvent	B	UINT32	The last clear shutdown sequential number of events. If there is no event function, the property will not be displayed.
SeqNumThisSessionBootEvent	B	UINT32	This session boot sequential number of events. If there is no event function, the property will not be displayed.

6.2 Method

Table 6.2-1

Name	Vendor Only	Description	Parameters D-bus Type	Response D-bus Type
GetEvents	B/M	Get Event Detail	UINT32 <SequenceNumber> UINT32 <SeqCount>	● ARRAY ○ STRUCT □ UINT32 SequenceNumber □ UINT32 EventCode: To index into Event Description □ STRING EventLevel · Debug · Progress · Infromation · Warning · Critical · Fatal · Dead □ STRING Time

Name	Vendor Only	Description	Parameters D-bus Type	Response D-bus Type
				□ STRING Description

6.3 Signal

There is no such implementation.

6.4 Example

- Get Event Sequence Information (Get properties of StorageControllerEvent)

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1/Events org.freedesktop.DBus.Properties.GetAll
string:"com.insyde.storage.StorageControllerEvent"

Return Result:
   array [
     dict entry(
       string "SeqNumNewestEvent"
       variant          uint32 300
     )
     dict entry(
       string "SeqNumOldestEvent"
       variant          uint32 1
     )
     dict entry(
       string "SeqNumLastClearEvent"
       variant          uint32 0
     )
     dict entry(
       string "SeqNumLastCleanShutdownEvent"
       variant          uint32 0
     )
     dict entry(
       string "SeqNumThisSessionBootEvent"
       variant          uint32 0
     )
     dict entry(
       string "Result"
       variant          string "Success"
     )
   ]
```

- Get Event Details

```
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1/Events
com.insyde.storage.StorageControllerEvent.GetEvents uint32:298 uint32:2

Rerurn Result:
   array [
     struct {
       uint32 298
```

```
      uint32 65536
      string "Information"
      string "12/24/2024 ; 9:52:15"
      string "Hot-plug drive removed, Port=C0 Box=1 Bay=1 SN=2324E6E2DECA        "
    }
    struct {
      uint32 299
      uint32 262144
      string "Information"
      string "12/24/2024 ; 9:52:15"
      string "Physical drive failure, Port=C0 Box=1 Bay=1 reason=0x14"
    }
  ]
  array [
    dict entry(
      string "Result"
      variant            string "Success"
    )
  ]
```

7 Drive

A physical drive, which contains model, serial number and so on. For <DRIVE_ID> in Object Path, please refer to the value of property "StorageDrives" in Storage.

- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/Drives/<DRIVE_ID>
- Interface: com.insyde.storage.Drive

7.1 Properties – Normal

Table 7.1-1

Name	Vendor Only	D-Bus Type	Description
Description	B/M	STRING	Description of the drive.
Name	B/M	STRING	The name of the drive
SerialNumber	B/M	STRING	The serial number of the drive.
Model	B/M	STRING	The model number of the drive.
Revision	B/M	STRING	The revision of the drive. This is typically the firmware/hardware version of the drive.
HotspareType	B/M	STRING	The type of hot-spare, this drive is currently serving as. Valid values: • **None**, the drive is not currently a hot-spare. • **Global**, the drive is currently serving as a hot-spare for all other drives in the storage system. • **Dedicated**, the drive is currently serving as a hot-spare for a user defined set of drives.
EncryptionAbility	B/M	STRING	The encryption abilities of this drive. Valid Values: • **None**, the drive is not capable of self-encryption.
EncryptionStatus	B/M	STRING	The status of the encryption of this drive. Valid Values: • **Unencrypted**, the drive is not currently encrypted.
CapacityBytes	B/M	UINT64	The size in bytes of the drive.
Protocol	B/M	STRING	The protocol this drive is using to communicate to the storage controller. Valid Values: • SCSI • SAS • SATA • Fiber • NVME • AHCI
MediaType	B/M	STRING	The type of media contained in this drive. Valid Values: • **HDD**, the drive media type is traditional magnetic platters. • **SSD**, the drive media type is solid state or flash memory.

Name	Vendor Only	D-Bus Type	Description
CapableSpeedGbs	B/M	DOUBLE	The maximum speed supported by the device.
NegotiatedSpeedGbs	B/M	DOUBLE	The negotiated link speed for data transfer to and from the physical drive.
BlockSizeBytes	B/M	INT32	The size of each block. Unit in byte.
Volumes	B/M	ARRAY <STRING>	Volume IDs.
Status.State	B/M	STRING	This indicates the known state of the resource, such as if it is enabled. Valid values: • **Enabled**, this function or resource has been enabled. • **Disabled**, this function or resource has been disabled. • **StandbyOffline**, this function or resource is enabled,but awaiting an external action to activate it. • **StandbySpare**, this function or resource is part of a redundancy set and is awaiting a failover or other external action to activate it. • **UnavailableOffline**, this function or resource is present but cannot be used. • **Updating**, the element is updating and may be unavailable or degraded. • **Absent**, this function or resource is not present or not detected.
Status.Health	B/M	STRING	This represents the health state of this resource in the absence of its dependent resources. Valid Values: • **OK**, Normal. • **Warning**, A condition exists that requires attention. • **Critical**, A critical condition exists that requires immediate attention.
Link.Oem.Controller	B/M	STRING	Controller Id.
Operations.PatrolReadProgressPercent	B	INT32	The percentage of the Patrol Read operation that has been completed.
Operations.CopyBackProgressPercent	B/M	INT32	The percentage of the Replace Drive operation that has been completed.
Oem.Temperature	B/M	INT32	The temperature of drive. Unit in Celsius.
Oem.SlotNumber	B	INT32	The slot number this drive inserts.
Oem.ConnectorIndex	B/M	INT32	The connector index of physical drive's active path port.
Oem.RPM	B	STRING	RPM of physical drive

7.2 Porperties - SMART for SATA Drive

The following chart lists some vendor-specific SMART attributes and the typical meaning of their raw values. The description below is a brief introduction of SMART attributes, for more information, please refer to Self-Monitoring, Analysis and Reporting Technology - Wikipedia.

Table 7.2-1

Name	Vendor Only	D-Bus Type	Description
Smart.ReadErrorRate	B/M	STRING	Stores data related to the rate of hardware read errors that occurred when reading data from a disk surface.
Smart.ThroughputPerformance	B/M	STRING	Overall (general) throughput performance of a hard disk drive.
Smart.Spin-UpTime	B/M	STRING	Average time of spindle spin up
Smart.Start/StopCount	B/M	STRING	A tally of spindle start/stop cycles.
Smart.ReallocatedSectorsCount	B/M	STRING	Count of reallocated sectors.
Smart.ReadChannelMargin	B/M	STRING	Margin of a channel while reading data.
Smart.SeekErrorRate	B/M	STRING	Rate of seek errors of the magnetic heads.
Smart.SeekTimePerformance	B/M	STRING	Average performance of seek operations of the magnetic heads.
Smart.Power-OnHours	B/M	STRING	Count of hours in power-on state.
Smart.SpinRetryCount	B/M	STRING	Count of retry of spin start attempts.
Smart.RecalibrationRetries	B/M	STRING	Count that recalibration was requested
Smart. PowerCycleCount	B/M	STRING	Count of full hard disk power on/off cycles.
Smart.SoftReadErrorRate	B/M	STRING	Uncorrected read errors reported to the operating system.
Smart.CurrentHeliumLevel	B/M	STRING	measures the helium inside of the drive specific to this manufacturer.

Smart.HeliumConditionLower	B/M	STRING	measures level of helium inside of the drive specific to this manufacturer.
Smart.HeliumConditionUpper	B/M	STRING	measures level of helium inside of the drive specific to this manufacturer.
Smart.AvailableReservedSpace	B/M	STRING	available reserved space as a percentage of the initial reserved space.
Smart.SSDProgramFailCount	B/M	STRING	The total number of flash program operation failures since the drive was deployed.
Smart.SSDEraseFailCount	B/M	STRING	Counts the number of flash erase failures.
Smart.SSDWearLevelingCount	B/M	STRING	Counts the maximum worst erase count on any block.
Smart.UnexpectedPowerLossCount	B/M	STRING	Raw value reports the number of unclean shutdowns, cumulative over the life of an SSD.
Smart.PowerLossProtectionFailure	B/M	STRING	Last test result as microseconds to discharge cap, saturated at its maximum value.
Smart.EraseFailCount	B/M	STRING	S.M.A.R.T. parameter indicates a number of flash erase command failures.
Smart.WearRangeDelta	B/M	STRING	Delta between most-worn and least-worn Flash blocks.
Smart.UsedReservedBlockCount	B/M	STRING	"Pre-Fail" attribute used at least in Samsung devices.
Smart.UsedReservedBlockCountTotal	B/M	STRING	"Pre-Fail" attribute used at least in Samsung devices.
Smart.UnusedReservedBlockCountTotal	B/M	STRING	"Pre-Fail" attribute used at least in HP devices.
Smart.Program FailCount Total	B/M	STRING	Total number of Flash program operation failures.
Smart.EraseFailCount	B/M	STRING	"Pre-Fail" Attribute used at least in Samsung devices.

Smart.SATADownshiftErrorCount	B/M	STRING	uncorrectable errors encountered during normal operation.
Smart. End-to-Enderror/IOEDC	B/M	STRING	count of parity errors which occur in the data path to the media via the drive's cache RAM.
Smart. HeadStability	B/M	STRING	Western Digital attribute.
Smart. InducedOp-VibrationDetection	B/M	STRING	Western Digital attribute.
Smart. ReportedUncorrectableErrors	B/M	STRING	The count of errors that could not be recovered using hardware ECC.
Smart. CommandTimeout	B/M	STRING	The count of aborted operations due to HDD timeout.
Smart. HighFlyWrites	B/M	STRING	count of errors when a recording head is flying outside its normal operating range. over the lifetime of the drive.
Smart. TemperatureDifference	B/M	STRING	Value is equal to (100-temp. °C), allowing manufacturer to set a minimum threshold which corresponds to a maximum temperature.
Smart. G-senseErrorRate	B/M	STRING	The count of errors resulting from externally induced shock and vibration.
Smart. Power-offRetractCount	B/M	STRING	Number of power-off or emergency retract cycles.
Smart. Smart.LoadCycleCount	B/M	STRING	Count of load/unload cycles into head landing zone position.
Smart. Temperature or TemperatureCelsius	B/M	STRING	Indicates the device temperature.
Smart. HardwareECCRecovered	B/M	STRING	The raw value has different structure for different vendors.
Smart. ReallocationEventCount	B/M	STRING	Count of remap operations.
Smart. CurrentPendingSectorCount	B/M	STRING	Count of "unstable" sectors (waiting to be remapped

Smart. UncorrectableSectorCount	B/M	STRING	The total count of uncorrectable errors when reading/writing a sector.
Smart. UltraDMACRCErrorCount	B/M	STRING	The count of errors in data transfer via the interface cable as determined by ICRC.
Smart. WriteErrorRate	B/M	STRING	The total count of errors when writing a sector.
Smart. SoftReadErrorRate	B/M	STRING	Count indicates the number of uncorrectable software read errors.
Smart. DataAddressMarkerrors	B/M	STRING	Count of Data Address Mark errors (or vendor-specific).
Smart. RunOutCancel	B/M	STRING	The number of errors caused by incorrect checksum during the error correction.
Smart. SoftECCCorrection	B/M	STRING	Count of errors corrected by the internal error correction software.
Smart. ThermalAsperityRate	B/M	STRING	Count of errors due to high temperature.
Smart. FlyingHeight	B/M	STRING	Height of heads above the disk surface.
Smart. SpinHighCurrent	B/M	STRING	Amount of surge current used to spin up the drive.
Smart. SpinBuzz	B/M	STRING	Count of buzz routines needed to spin up the drive due to insufficient power.
Smart. OfflineSeekPerformance	B/M	STRING	Drive's seek performance during its internal tests.
Smart. VibrationDuringWrite	B/M	STRING	Found in Maxtor 6B200M0 200GB and 2R015H1 15GB disks.
Smart. VibrationDuringWrite	B/M	STRING	A recording of a vibration encountered during write operations.
Smart. ShockDuringWrite	B/M	STRING	A recording of shock encountered during write operations.
Smart. DiskShift	B/M	STRING	Distance the disk has shifted relative to the spindle.

Smart. G-SenseErrorRate	B/M	STRING	The count of errors resulting from externally induced shock and vibration.
Smart.LoadedHours	B/M	STRING	Time spent operating under data load.
Smart.Load/UnloadRetryCount	B/M	STRING	Count of times head changes position.
Smart.LoadFriction	B/M	STRING	Resistance caused by friction in mechanical parts while operating.
Smart.Load/UnloadCycleCount	B/M	STRING	Total count of load cycles Some drives use 193 (0xC1) for Load Cycle Count instead.
Smart.Load'In'-time	B/M	STRING	Total time of loading on the magnetic heads actuator.
Smart.TorqueAmplificationCount	B/M	STRING	Count of attempts to compensate for platter speed variations.
Smart.Power-OffRetractCycle	B/M	STRING	The number of power-off cycles which are counted whenever there is a "retract event”.
Smart.GMRHeadAmplitude	B/M	STRING	Amplitude of "thrashing".
Smart.LifeLeft	B/M	STRING	Indicates the approximate SSD life left, in terms of program/erase cycles or available reserved blocks.
Smart.EnduranceRemaining	B/M	STRING	Number of physical erase cycles completed on the SSD as a percentage of the maximum physical erase cycles the drive is designed to endure.
Smart.MediaWearoutIndicator	B/M	STRING	occasionally used for Power-On Hours
Smart.AverageerasecountANDMaximumEraseCount	B/M	STRING	Decoded as: byte 0-1-2=average erase count (big endian) and byte 3-4-5=max erase count (big endian).
Smart.GoodBlockCountANDSystem(Free)BlockCount	B/M	STRING	Decoded as: byte 0-1-2=good block count (big endian) and byte 3-4=system (free) block count.

Smart.HeadFlyingHours	B/M	STRING	Time spent during the positioning of the drive heads. Some Fujitsu drives report the count of link resets during a data transfer.
Smart.TotalLBAsWritten	B/M	STRING	Total count of LBAs written.
Smart.TotalLBAsRead	B/M	STRING	Total count of LBAs read.
Smart.TotalLBAsWrittenExpanded	B/M	STRING	The upper 5 bytes of the 12-byte total number of LBAs written to the device
Smart.TotalLBAsReadExpanded	B/M	STRING	The upper 5 bytes of the 12-byte total number of LBAs read from the device.
Smart.RemainingRatedWriteEndurance	B/M	STRING	Certified Dell SSDs use this for write endurance.
Smart.CumulativeHostSectorsWritten	B/M	STRING	(Micron) LBA written due to computer request.
Smart.HostProgramPageCount	B/M	STRING	(Micron) NAND pages written due to computer request.
Smart.BackgroundProgramPageCount	B/M	STRING	(Micron) NAND pages written due to background operations (e.g. garbage collection).
Smart.NANDWrites(1GiB)	B/M	STRING	Total NAND Writes. Raw value reports the number of writes to NAND in 1 GB increments.
Smart.ReadErrorRetryRate	B/M	STRING	Count of errors while reading from a disk.
Smart.MinimumSparesRemaining	B/M	STRING	The Minimum Spares Remaining attribute indicates the number of remaining spare blocks as a percentage of the total number of spare blocks available.[
Smart.NewlyAddedBadFlashBlock	B/M	STRING	The Newly Added Bad Flash Block attribute indicates the total number of bad flash blocks the drive detected since it was first initialized in manufacturing.
Smart.FreeFallProtection	B/M	STRING	Count of "Free Fall Events" detected.

7.3 Properties - SMART for NVMe Drive

The description below is a brief introduction of SMART attributes, for more information, please refer to NVM Express® Base Specification, SMART / Health Information.

Table 7.3-1

Name	Vendor Only	D-Bus Type	Description
Smart.CriticalWarning	B/M	INT	Critical warnings for the state of the controller.
Smart.CompositeTemperature	B/M	INT	Temperature in Kelvins.
Smart.AvailableSpare	B/M	INT	Percentage of available spare space (spare space for bad block mapping).
Smart.AvailableSpareThreshold	B/M	INT	Percentage of available spare space threshold.
Smart.PercentageUsed	B/M	INT	Percentage of drive life used. This drive life may be estimated as terabytes written.
Smart.DataUnitsRead	B/M	INT	Number of 512-byte data units the host has read from the controller. This value does not include metadata. This value is reported in thousands (i.e., a value of 1 corresponds to 1000 units of 512 bytes written) and is rounded up.
Smart.DataUnitsWritten	B/M	INT	Number of 512-byte data units the host has written to the controller. This value does not include metadata. This value is reported in thousands (i.e., a value of 1 corresponds to 1000 units of 512 bytes written) and is rounded up.
Smart.HostReadCommands	B/M	INT	Number of read commands completed by the controller.
Smart.HostWriteCommands	B/M	INT	Number of write commands completed by the controller.
Smart.ControllerBusyTime	B/M	INT	Amount of time the controller is busy with I/O commands.
Smart.PowerCycles	B/M	INT	Number of power cycles.
Smart.PowerOnHours	B/M	INT	Number of power-on hours, excluding time powered on in non-operational power state.
Smart.UnsafeShutdowns	B/M	INT	Number of unsafe shutdowns. Incremented when a Shutdown Notification is not received prior to loss of power.
Smart.MediaDataIntegrityHours	B/M	INT	Number of occurrences where the controller detected an unrecovered data integrity error, including uncorrectable ECC, CRC checksum failure, or LBA tag mismatch.
Smart.NumErrorLogEntries	B/M	INT	Number of Error Information log entries over the life of the controller.

Smart.WarningCompositeTempTime	B/M	INT	The amount of time in minutes that the controller is operational and the Composite Temperature is greater than or equal to the Warning Composite Temperature Threshold and less than the Critical Composite Temperature Threshold.
Smart.CriticalCompositeTempTime	B/M	INT	Contains the amount of time in minutes that the controller is operational and the Composite Temperature is greater than or equal to the Critical Composite Temperature Threshold.
Smart.TemperatureSensor1	B/M	INT	
Smart.TemperatureSensor2	B/M	INT	
Smart.TemperatureSensor3	B/M	INT	
Smart.TemperatureSensor4	B/M	INT	
Smart.TemperatureSensor5	B/M	INT	
Smart.TemperatureSensor6	B/M	INT	
Smart.TemperatureSensor7	B/M	INT	
Smart.TemperatureSensor8	B/M	INT	
Smart.ThermalMgmtTemp1Transitions	B/M	INT	
Smart.ThermalMgmtTemp2Transitions	B/M	INT	
Smart.TotalTimeThermalMgmTemp1	B/M	INT	

Smart.TotalTimeThermalMgmTemp2	B/M	INT	

7.4 Method

Table 7.3-1

Name	Vendor Only	Description	Parameters D-bus Type	Response D-bus Type
MakeDedicatedHotspare	B/M	Make a dedicated hot spare drive for a specific logical drive. Note: Does not support dedicated hot spare for RAID0.	STRING <VolumeId>	• STRING ○ Success ○ Failure
MakeGlobalHotspare	B/M	Make a global hot spare drive for all virtual drives.	N/A	• STRING ○ Success ○ Failure
RemoveHotspare	B/M	Remove a hot spare drive.	N/A	• STRING ○ Success ○ Failure
StartReplacePD	B/M	Replace a physical drive with another one.	STRING <Destination DriveID>	• STRING ○ Success ○ Failure
StopReplacePD	B	Stop to Replace a drive	N/A	• STRING ○ Success ○ Failure
StartLocatePD	B/M	Indicate LED of specific Physical Drive. Note: For Microchip, the LED will be turned off after 7 days.	N/A	• STRING ○ Success ○ Failure
StopLocatePD	B/M	Stop to indicate LED of specific Physical Drive	N/A	• STRING ○ Success ○ Failure

7.5 Signal

There is no such implementation.

7.6 Example

• Get Physical Drive Information

```
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/0
org.freedesktop.DBus.Properties.GetAll string:"com.insyde.storage.Drive"

Return Results:
   array [
```

```
    dict entry(
      string "Description"
      variant          string "Physical Disk 0"
    )
    dict entry(
      string "Name"
      variant          string "Physical Disk 0"
    )
    dict entry(
      string "SerialNumber"
      variant          string "2324E6E27247"
    )
    dict entry(
      string "Model"
      variant          string "ATA     CT500MX500SSD1"
    )
...
```

● Hot Spare

```
[Make Dedicated Hot Spare] Make a dedicated hotspare drive id 2 for logical drive "Broadcom0_1"
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.MakeDedicatedHotspare string:"Broadcom0_1"

ps. RAID-0 does not support dedicated hotspare.

[Make Global Hot Spare] Make a global hotspare drive id 2
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.MakeGlobalHotspare

[Remove Hot Spare] Remove a hotspare drive id 2
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.RemoveHotspare
```

● Locate Physical Drive

```
[Locate Physical Drive] Locate physical drive id 2
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.StartLocatePD

[Stop Locate Physical Drive] Stop to locate physical drive id 2
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.StopLocatePD
```

● Start Replace Physical Drive

```
[Replace physical drive 0 with physical drive 2]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/0
com.insyde.storage.Drive.StartReplacePD string:"2"

ps. Environment: At least three hard drives. Drive index 0 and 1 create RAID1. Start replacing the destination drive (index 2)
with the source drive (index 0)
```

● Stop Replace Physical Drive - For Broadcom RAID

```
[Stop physical drive 2 from replacing other physical drive]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Drives/2
com.insyde.storage.Drive.StopReplacePD

ps. Environment: At least three hard drives. Drive index 0 and 1 create RAID1. Stop the drive (index 2) that is doing replace from
a source drive.
```

8 VolumeCollection

A collection of logical drives (aka volumes), which contains count and members.
- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/Volumes
- Interface: com.insyde.storage.VolumeCollection

8.1 Properties

Table 8.1-1

Name	Vendor Only	D-bus Type	Description
Description	B/M	STRING	A group of logical drives.
Name	B/M	STRING	The name : "Volumes Collection".
Count	B/M	INT32	The logical drive count.
Members	B/M	ARRAY <STRING>	The logical drive IDs.

8.2 Method

There is no such implementation.

8.3 Signal

There is no such implementation.

8.4 Example

- Get VolumeCollection Information

```
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes
org.freedesktop.DBus.Properties.GetAll string:"com.insyde.storage.VolumeCollection"

Return Result:
   array [
      dict entry(
         string "Description"
         variant          string "Volumes Disk"
      )
      dict entry(
         string "Name"
         variant          string "Volumes Collection"
      )
      dict entry(
         string "Count"
         variant          int32 1
      )
      dict entry(
         string "Members"
         variant          array [
               string "Broadcom0_1"      <------ logical drive id
            ]
```

```
    )
    dict entry(
      string "Result"
      variant          string "Success"
    )
  ]
```

9 Volume

A logical drive (aka volume), which contains Name, Capacity, Type and so on. For <VOLUME_ID> in Object Path, please refer to the value of property "Members" in VolumeCollection.

- D-bus Name: com.insyde.storage
- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/Volumes/<VOLUME_ID>
- Interface: com.insyde.storage.Volume

9.1 Properties

Table 9.1-1

Name	Vendor Only	D-bus Type	Description
Description	B/M	STRING	Description about the Volume.
Name	B/M	STRING	The name of the Volume.
CapacityBytes	B/M	UINT64	The size in bytes of this Volume.
VolumeType	B/M	STRING	The type of this volume. Valid Values: • **RawDevice**, the volume is a raw physical device without any RAID or other virtualization applied. • **NonRedundant**, the volume is a non-redundant storage device. • **Mirrored**, the volume is a mirrored device. • **StripedWithParity**, the volume is a device which uses parity to retain redundant information.
BlockSizeBytes	B/M	INT32	The size of block. Unit in byte.
OptimumIOSizeBytes	B	INT32	The size in bytes of this Volume's optimum IO size.
Drives	B/M	ARRAY <STRING>	Physical drive IDs.
Status.State	B/M	STRING	This indicates the known state of the resource, such as if it is enabled. Valid values: For Broadcom: • **Enabled**, this function or resource has been enabled. • **UnavailableOffline**, this function or resource is present but cannot be used. For Microchip: • **OK**, The logical device is healthy and is readily accessible by the host. • **Fail**, The logical device has bad or missing physical device(s). • **Interim Recovery**, The logical device has a bad or missing drive. Logical device is operating with reduced performance and a further physical drive failure may result in data loss depending on the fault tolerance. To correct this problem, check the data and power connections to the physical drives or replace the failed drive. • **Ready For Recovery**, The logical device is queued to be recovered from a failed physical device.

Name	Vendor Only	D-bus Type	Description
			• **Recovering**, The logical device is rebuilding a physical device from fault tolerant data. • **Wrong Drive Replaced**, A wrong physical device was replaced. • **Drive Improperly Connected**, A physical device is not properly connected. • **Hardware Is Overheating**, A physical device temperature is about to reach the threshold value. • **Hardware Has Overheated**, A physical device temperature has crossed the threshold value. • **Expanding**, The logical device is undergoing online capacity expansion. • **Not Yet Available**, An expand, shrink, or move operation on the array is in progress. This logical device will remain in this state until all expand, shrink, or move operations on this array are completed. All I/O requests sent to the logical drive in this state will be rejected. • **Queued For Expansion**, The logical device is queued for expansion. • **Disabled From SCSI ID Conflict**, A conflict with an existing SCSI ID exists. Check all SCSI components to make sure they all have a unique SCSI ID. • **Ejected**, The logical device is offline from being ejected. Reinstall the removed physical devices. • **Erase In Progress**, The logical device is offline and has erase in progress. • **Unused**, Logical Drive status unused. • **Ready To Perform Predictive Spare Rebuild**, • **RPI In Progress**, Rapid parity initialization is currently in progress on this logical device. • **RPI Queued**, Rapid parity initialization is currently queued on this logical device. It will start once other progress tasks are completed. • **Encrypted Volume Without Key**, The encrypted logical device is exported from a foreign controller with different master key. Please import the foreign master key to access the logical device. • **Plaintext Volume Rejected In Encrypting Mode**, Logical device is plaintext and cannot be accessed as the controller is in encryption-only mode. • **Encryption Migration**, The logical device is being migrated between plaintext and ciphertext. • **Encrypted Volume Rekeying**, The logical device is encrypted and all data is being re-keyed using the background 'online capacity expansion' transformation task. The cache memory is being used to keep track of progress. • **Encrypted Volume Encryption Off**, Logical Drive status encrypted with encryption off. • **Volume Encode Requested**, Logical Drive status volume encode requested. • **Encrypted Volume Rekey Requested**, The logical device is encrypted and has received a request to re-key all data with a new encryption key. • **Unsupported On This Controller**, Logical device is

Name	Vendor Only	D-bus Type	Description
			unsupported on this controller. Host access to this volume is denied. Logical device can still be deletd/reconfigured with data loss.
Status.Health	B/M	STRING	This represents the health state of this resource in the absence of its dependent resources. Valid Values: • **OK**, Normal. • **Warning**, A condition exists that requires attention. • **Critical**, A critical condition exists that requires immediate attention.
Oem.RaidLevel	B/M	STRING	The raid level of the volume.
Oem.State	B/M	STRING	The state of the volume. Valid values: For Broadcom: • **Optimal**, A virtual drive whose members are all online. • **Partially Degraded**, A virtual drive with a redundant RAID level that is capable of sustaining more than one member drive failure. This state also applies to the virtual drive's member drives. Currently, a RAID 6 or RAID 60 virtual drive is the only virtual drive that can be partially degraded. • **Degraded**, A virtual drive with a redundant RAID level with one or more member failures and can no longer sustain a subsequent drive failure. • **Offline**, A virtual drive with one or more member failures that make the data inaccessible. For Microchip: please refer to "Status.State"
Oem.ReadPolicy	B	STRING	The read policy of the volume. Valid value: • **Read Ahead**, Permits the controller to read sequentially ahead of the requested data and allows the controller to store the additional data in the cache memory. Here, the controller anticipates that the data is required frequently. Even though Always Read Ahead policy speeds up the reads for sequential data, but little improvement is seen when accessing the random data. • **No Read Ahead**, Disables the Always Read Ahead capability of the controller
Oem.WritePolicy	B	STRING	The write policy of the volume. Valid value: • **Write Back** – In this mode, the controller sends a data transfer completion signal to the host when the controller cache receives all of the data in a transaction. If you select the Write Back policy and the battery is absent, the firmware disables the Write Back policy and defaults to the Write Through policy. • **Write Through** – In this mode, the controller sends a data transfer completion signal to the host when the drive subsystem receives all the data in a transaction. • **Write Cached** – In this mode, the controller sends a data transfer completion signal to the host when the controller cache receives all the data in a transaction. If you select the Write Cache policy and the battery is bad, the firmware is forced to use the Write Back policy.
Oem.IoPolicy	B	STRING	The IO policy of the volume. Valid value: • **Cached IO**, In this mode, all reads are buffered in cache memory. Cached I/O provides faster processing. • **Direct IO**, In this mode, reads are not buffered in cache memory. Data is transferred to the cache and the host concurrently. If the same data block is read again, it comes from cache memory. Direct I/O makes sure that the cache

Name	Vendor Only	D-bus Type	Description
			and the host contain the same data. (The IO Policy applies to read operations on a specific virtual drive. It does not affect the read-ahead cache.)
Oem.AccessPolicy	B	STRING	The access policy of the virtual drive. Valid value: • **Read Write**, Allow read/write access. This setting is the default value. • **Read Only**, Allow read-only access. • **Blocked**, Do not allow access.
Oem.DiskCachePolicy	B	STRING	The disk cache policy of the volume. Valid value: • **Enable**, Enable the drive cache. • **Disable**, Disable the drive cache. • **Unchanged**, Updating the drive cache policy to Unchanged may enable /disable the drive cache based on the WCE (Write Cache Policy) bit of the save mode page of the drive.
Oem.StripSize	B/M	INT	The strip size (KB) of the volume.
Oem.BadBlocks	B	BOOLEAN	Check whether the volume has bad blocks.
Oem.BGInit	B	BOOLEAN	Check whether the volume does Background Initialization.
Oem.Accelerator	M	STRING	Logical Drive Accelerator. Valid values: • **Controller Cache**, Caching for logical device enabled through controller cache. • **None**, no cache
Oem.InitMethod	M	STRING	The initialization method of the volume. Valid values: • **Default** • **Rapid Parity Initialization**, Logical device is undergoing the Rapid Parity initialization and may not available until it is completed. (For RAID5,6,50 and 60)
Oem.BackgroundConsistencyCheckStatus	M	STRING	The background consistency check status for the volume. Valid values: • **Complete**, surface analysis consistency initialization has completed on the logical drive. • **Failed**, surface analysis consistency check failed. • **Not Started**, surface analysis has not started on the logical drive because it has not been written to yet. • **Scanning**, the logical drive is currently being scanned.
Operations.CheckConsistencyProgressPercent	B	INT32	The percentage of the CheckConsistency operation that has been completed.
Operations.InitializationProgressPercent	B	INT32	The percentage of the Initialization operation that has been completed.
Operations.ReConstructionProgressPercent	B/M	INT32	The percentage of the ReConstruction operation that has been completed.
Link.Oem.Controller	B/M	STRING	Controller ID.

9.2 Method

Table 9.2-1

Name	Vendor Only	Description	Parameters D-bus Type	Response D-Bus Type
CheckConsistency	B	Check consistency for a specific logical drive.	N/A	• STRING ○ Success ○ Failure
CancelCheckConsistency	B	Cancel consistency check.	N/A	• STRING ○ Success

Name	Vendor Only	Description	Parameters D-bus Type	Response D-Bus Type
				○ Failure
StartLocateLD	B	Indicate LED of specific logical drive.	N/A	• STRING ○ Success ○ Failure
StopLocateLD	B	Stop to indicate LED of specific logical drive.	N/A	• STRING ○ Success ○ Failure
ChangeLDConfig	B/M	Change a Logical Drive Configuration: migrating the RAID level for a logical drive. (aka perform Reconstrunction)	• STRING RaidLevel • ARRAY <STRING SelectedDriveIds>	• STRING ○ Success ○ Failure

9.3 Signal

There is no such implementation.

9.4 Properties Set

- Object Path: /com/insyde/storage/Systems/<SYSTEMS_SN>/Storage/<STORAGE_ID>/Volumes/<VOLUME_ID>
- Interface: org.freedesktop.DBus.Properties
- Method: Set

Table 9.4-1

Feature	Vendor only	Description	Parameters D-bus Type	Response D-bus Type
Rename	B	Modify the name of the logical drive	• STRING "com.insyde.storage.Volume" • STRING "Rename" • STRING <NewName>	N/A
Read Policy	B	Read policy for the logical drive. For more details, please read the description of Oem.ReadPolicy.	• STRING "com.insyde.storage.Volume" • STRING "Read Policy" • STRING <Value> Ex. Read Ahead No Read Ahead	N/A
Write Policy	B	Write policy for the logical drive. For more details, please read the description of Oem.WritePolicy.	• STRING "com.insyde.storage.Volume" • STRING "Write Policy" • STRING <Value> Ex. Write Through Write Back Write Cached	N/A
IO Policy	B	IO Policy for the logical drive. For more details, please read the description of Oem.IOPolicy.	• STRING "com.insyde.storage.Volume" • STRING "IO Policy" • STRING <Value> Ex.	N/A

			Direct IO Cached IO	
Disk Cache Policy	B	Disk cache policy for the logical drive. For more details, please read the description of Oem.DiskCachePolicy.	• STRING "com.insyde.storage.Volume" • STRING "Disk Cache Policy" • STRING <Value> Ex. Enable Disable Unchanged	N/A
BGI	B	When BGI (Background Initialization) is enabled, the firmware runs the initialization process in the background. When BGI is disabled, the initialization process does not start automatically and does not run in the background.	• STRING "com.insyde.storage.Volume" • STRING "BG Init" • BOOLEAN <Value> Ex. true false	N/A
Accelerator	M	Logical Device Acceleration Method indicates whether caching for logical device enabled through controller cache.	• STRING "com.insyde.storage.Volume" • STRING "MC LD Accelerator" • STRING <Value> Ex. Controller Cache None	N/A

9.5 Example

• Get Logical Drive Information

```
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes/
Broadcom239_1 org.freedesktop.DBus.Properties.GetAll string:"com.insyde.storage.Volume"

Rerutn Result:
  array [
    dict entry(
      string "Description"
      variant          string "Logical Disk Broadcom239_1"
    )
    dict entry(
      string "Name"
      variant          string "test"
    )
    dict entry(
      string "CapacityBytes"
      variant          uint64 499558383616
    )
...
    dict entry(
      string "Drives"
      variant          array [
          string "0"
          string "1"
        ]
    )
    dict entry(
      string "Status.State"
      variant          string "Enabled"
```

```
    )
    dict entry(
      string "Status.Health"
      variant          string "OK"
    )
    dict entry(
      string "Oem.RaidLevel"
      variant          string "1"
    )
...
```

● Set Logical Drive Properties - For Broadcom RAID

```
[Rename]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Rename" variant:string:"test1"

[Read Policy]
[[Read Ahead]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Read Policy" variant:string:"Read Ahead"
[[No Read Ahead]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Read Policy" variant:string:"No Read Ahead"


[Write Policy]
[[Write Through]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Write Policy" variant:string:"Write Through"
[[Write Back]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Write Policy" variant:string:"Write Back"
[[Write Cached]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Write Policy" variant:string:"Write Cached"

[IO Policy]
[[Direct IO]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"IO Policy" variant:string:"Direct IO"
[[Cached IO]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"IO Policy" variant:string:"Cached IO"

[Disk Cache Policy]
[[Enable]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Disk Cache Policy" variant:string:"Enable"
[[Disable]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
```

```
string:"com.insyde.storage.Volume" string:"Disk Cache Policy" variant:string:"Disable"
[[Unchanged]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"Disk Cache Policy" variant:string:"Unchanged"

[BGI]
[[true]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"BG Init" variant:boolean:true
[[false]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"BG Init" variant:boolean:false
```

● Set Logical Drive Properties - For Microchip RAID (MCTP over PCIe)

```
[Logical Drive Accelerator]
[[Controller Cache]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/Volumes/Microsemi10 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"MC LD Accelerator" variant:string:"Controller Cache"
[[None]]
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/Volumes/Microsemi10 org.freedesktop.DBus.Properties.Set
string:"com.insyde.storage.Volume" string:"MC LD Accelerator" variant:string:"None"
```

● Check Consistency - For Broadcom RAID

```
[Check Consistency]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes/
Broadcom239_1 com.insyde.storage.Volume.CheckConsistency
[Cancel Check Consistency]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes/
Broadcom239_1 com.insyde.storage.Volume.CancelCheck
```

● Locate Logical Drive - For Broadcom RAID

```
[Locate Logical Drive]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes/
Broadcom239_1 com.insyde.storage.Volume.StartLocateLD
[Stop Locate Logical Drive]
# dbus-send --system --print-reply --dest=com.insyde.storage /com/insyde/storage/Systems/123456789/Storage/1/Volumes/
Broadcom239_1 com.insyde.storage.Volume.StopLocateLD
```

● Perform Reconstruction (Change the RAID level of the logical drive)

```
Ex. Change logical drive "Broadcom239_1" RAID level from 0 to 1
Step1: Create RAID1 first
Step2: Change RAID Level to 0 by following command
# dbus-send --system --print-reply --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789/Storage/1/Volumes/Broadcom239_1 com.insyde.storage.Volume.ChangeLDConfig
string:"0"
```

10 Reference

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